

AI-Enabled Teacher Assistance Program for CBSE Schools

Objective

To empower teachers with AI-based tools that simplify lesson planning, enhance teaching delivery, and personalize student learning — aligned with the **NCERT/CBSE curriculum**.

Program Structure

Phase 1: AI Assistance for Teachers (Teaching & Content Delivery Support)

This phase focuses on **real-time teaching support** and **automated content generation**, enabling teachers to save time and deliver high-quality, engaging lessons.

1. Lesson Planning and Overview Generation

- **Input:** Subject, Class, Chapter/Topic
 - **AI Output:**
 - Generates **lesson overview** as per NCERT syllabus.
 - Includes **key learning outcomes, important definitions, formulae, and summary points**.
 - Provides **dual-mode delivery**:
 - **Text Format:** Readable like a teacher's hand-written board notes.
 - **Speech Format:** Slow-paced narration with clear pronunciation — as if a teacher is writing and explaining on a board.
-

2. Interactive Teaching via AI-Powered PPT

- Auto-generates or loads **pre-designed PowerPoint slides** for each topic.
 - Slides include:
 - **Graphics and diagrams**
 - **Step-by-step explanations**
 - **Key terms highlighted** for classroom teaching.
 - Teacher can **customize slides** or directly present them through classroom displays.
-

3. On-Demand Doubt Resolution

- When students raise doubts on **specific subtopics**, AI instantly:
 - Generates **simplified explanations** with examples.
 - Optionally supports **bilingual mode (English + Hindi)**.
 - Allows teacher to project the AI response for classroom explanation.
-

4. Animated & Activity-Based Learning

- Teachers can request **preloaded animation videos** that visually explain the concept.
 - **Activity Videos** are also available to encourage **hands-on learning** or **real-life applications** of the topic.
 - Both are **curriculum-tagged** and age-appropriate.
-

5. Recorded Lecture Assistance

- If the teacher wants to play an **existing recorded lecture**,
 - The AI **LLM searches** within the database of recorded videos.
 - Finds the **most relevant section** of the lecture related to the topic.
 - Plays the video seamlessly without manual searching.
-

6. Automated Classwork Generation

- After each lecture, AI generates **topic-specific classwork** in multiple formats:
 - **MCQs** (concept-based)
 - **One-liners**
 - **Short questions with answers**
 - Teachers can **print or display** these for classroom assessment.
-

7. Homework & Assignment Creation

- AI generates **customized homework sheets**:
 - **Short answer** and **long answer questions**.
 - Aligned with the topic covered in class.
 - Teachers can **adjust difficulty level** (easy, moderate, advanced).
 - Homework sheets are automatically formatted for **print or digital submission**.
-

8. Save Lecture Notes

- Post completion of teaching, ppt would be saved as PDF and ready to share with students
- This ppt will work as a class notes for students for future assistance.

9. Next-Day Homework Explanation

- On the following day, AI provides:
 - **Step-by-step explanation** for the previous day's homework.
 - **Audio-visual support** for difficult questions.
 - Option for **student-wise feedback** based on submission (in later phases).

10. Generating Mind maps and flash cards

- Generates Mind Maps and flash card for quick learning of students
- Generates flow charts
- Important questions based on PYQs

Technology Stack (Phase 1)

Component	Description
LLM (AI Engine)	Custom-trained on CBSE syllabus, NCERT content, and educational pedagogy.
Speech & Animation Engine	Converts text to slow-motion speech + simulates board-writing animations.
Content Repository	Preloaded PPTs, recorded videos, animation clips, and activity content tagged topic-wise.
Teacher Dashboard	Web-based UI for lesson planning, content access, and homework generation.
Student Interface (Limited)	For accessing homework, activity videos, and classwork review.

Phase 2: AI Evaluation, Feedback & Insight System

This phase extends the program into **assessment, analytics, and adaptive learning**, helping teachers understand each student’s learning pattern, strengths, and weaknesses.

1. AI-Based Evaluation System

Feature	Description
Objective Evaluation (Auto-grading)	Automatically evaluates MCQs, True/False, Fill in the blanks, and Match the following from classwork/homework.
Subjective Evaluation (AI-assisted)	Uses NLP and LLM to analyze short and long answers , checking concept coverage, grammar, and relevance.
Step-by-Step Evaluation (Math & Science)	For problem-solving subjects, AI checks solution steps , not just final answers, giving partial marks and hints.
Handwriting Recognition (Optional)	OCR-based reading of scanned answer sheets for automatic evaluation and teacher verification.
Instant Result Generation	Auto-generates results with marks, percentages, and concept-wise breakdown for each student.

2. Personalized Feedback Engine

Feature	Description
Concept Mastery Map	AI maps every question to a specific NCERT concept ID and tracks each student’s mastery level.
Strength & Weakness Report	Generates detailed insights for every student and topic (e.g., “Strong in Motion, weak in Laws of Force”).
Remedial Suggestions	AI recommends short videos, animations, or interactive quizzes to improve weak concepts.
Teacher Insight Dashboard	Displays class-level analytics — who needs attention, average scores, topic-wise progress, and difficulty trends.
Parent Summary Sheet	Automatically creates a weekly progress report for parents in simple language.

3. AI-Generated Question Bank

Feature	Description
Auto Question Generation	Generates new questions from any chapter or subtopic based on NCERT text.

Feature	Description
Difficulty Adjustment	Allows teacher to choose difficulty level — Easy / Moderate / Challenging .
Exam Pattern Mapping	Aligns generated questions with CBSE exam formats (1 mark, 2 marks, 3 marks, 5 marks).
Printable & Digital Formats	Supports PDF export for printed worksheets and online quiz mode for digital classes.

4. Progress & Analytics Dashboard

View Type	Description
Teacher View	Real-time classroom performance, average accuracy, attendance tracking, homework submission trends.
Student View	Visual learning report showing topic completion, quiz history, and recommended videos for revision.
School Admin View	Institution-level analytics, class performance comparisons, teacher engagement metrics.

5. Predictive Learning Assistance

- AI predicts **student performance trends** using past data.
 - Suggests **revision plans** and **remedial classes** before exams.
 - Alerts teacher if a student's **concept retention or participation** drops.
 - Helps plan **custom tests or revision cycles** automatically.
-

6. Integration & Deployment

Component	Description
Integration with Phase 1	Phase 2 works seamlessly with lesson planning, homework, and recorded lecture tools.
Mobile App (Future)	Students can access AI feedback, revision tips, and performance reports on mobile.